

Clinical Reference Guide: Colon Cancer

Extracted Focus & Question-Answer Summary from Cancer.gov PDQ

DOCUMENT OVERVIEW

Focus Condition:	Colon Cancer
Semantic Group:	Disorders (T191)
Source Data:	National Cancer Institute (NCI) / CancerGov

Question 1: What is (are) Colon Cancer ?

Colon cancer is a disease in which malignant (cancer) cells form in the tissues of the colon. The colon is part of the body's digestive system, which removes and processes nutrients (vitamins, minerals, carbohydrates, fats, proteins, and water) from foods and helps pass waste material out of the body. The digestive system is made up of the esophagus, stomach, and small and large intestines. The colon (large bowel) is the first part of the large intestine and is about 5 feet long. Together, the rectum and anal canal make up the last part of the large intestine and are about 6 to 8 inches long, ending at the anus.

Question 2: Who is at risk for Colon Cancer? ?

An individual's personal and family health history affects the risk of developing colon cancer. Older age is a primary risk factor for most cancers, as the chance of getting cancer increases as you get older. Specific risk factors for colorectal cancer include:

- Having a family history of colon or rectal cancer in a first-degree relative (parent, sibling, or child).
- Having a personal history of cancer of the colon, rectum, or ovary.
- Having a personal history of high-risk adenomas (colorectal polyps measuring 1 centimeter or larger, or exhibiting abnormal cellular histology under a microscope).
- Having inherited changes in certain genes that increase the risk of familial adenomatous polyposis (FAP) or Lynch syndrome (hereditary nonpolyposis colorectal cancer).
- Having a personal history of chronic ulcerative colitis or Crohn's disease for 8 years or more.
- Drinking three or more alcoholic drinks per day.
- Smoking cigarettes.
- Being Black or African American.
- Being obese.

Question 3: What are the symptoms of Colon Cancer ?

Signs of colon cancer include blood in the stool or a persistent change in bowel habits. These and other symptoms may be caused by colon cancer or by other benign medical conditions. Patients should consult a doctor if they experience:

- A change in bowel habits, including diarrhea, constipation, or a feeling that the bowel does not empty all the way.
- Blood (either bright red or very dark) present in the stool.
- Stools that are narrower than usual.
- Frequent gas pains, bloating, fullness, or abdominal cramps.
- Weight loss for no known reason.
- Feeling very tired or lethargic.
- Vomiting.

Question 4: How to diagnose Colon Cancer ?

Specialized examinations of the colon and rectum, alongside laboratory and endoscopic evaluations, are used to detect and diagnose colon cancer:

- **Physical Exam and History:** An exam of the body to check general health signs, personal habits, past illnesses, and performing a clinical inspection for unusual lumps.
- **Digital Rectal Exam (DRE):** Insertion of a lubricated, gloved finger into the rectum to feel for lumps or abnormalities.
- **Fecal Occult Blood Test (FOBT):** A test to check solid waste for blood visible only under a microscope, which can indicate polyps, cancer, or other conditions. Includes *Guaiac FOBT* (using chemical discoloration) and *Immunochemical FOBT / FIT* (using antibodies to detect liquid blood samples).
- **Barium Enema (Lower GI Series):** A series of lower GI x-rays taken after liquid barium coats the rectal and colonic mucosa.
- **Sigmoidoscopy & Colonoscopy:** Insertion of thin, lighted endoscopes into the rectum and lower colon (sigmoidoscopy) or entire colon (colonoscopy) to visually inspect mucosa and excise polyps or tissue samples.
- **Virtual Colonoscopy (CT Colonography):** A procedure using computed tomography (CT) x-rays to create detailed 3D computer images showing polyps or colonic surface anomalies without sedation.
- **Biopsy:** The definitive diagnostic standard, involving microscopic examination of excised cellular tissues by a pathologist to confirm malignancy.

Question 5: What is the outlook for Colon Cancer ?

The prognosis (chance of recovery) and the determination of appropriate therapeutic pathways depend on several critical clinical factors:

- **Disease Stage:** Whether the cancer is confined to the inner mucosal lining, has penetrated through the colon wall, or has metastasized to lymph nodes or distant organs.
- **Complications:** Whether the tumor has obstructed (blocked) or perforated (made a hole in) the colon.
- **Surgical Margins:** Whether any residual cancer cells remain after surgical resection.
- **Disease Status:** Whether the malignancy is a newly diagnosed primary tumor or a recurrence.
- **Patient Profile:** The patient's general health and baseline physical condition.
- **Tumor Markers:** Pre-treatment blood levels of *carcinoembryonic antigen (CEA)*, a substance that may be elevated when cancer is present.

Question 6: What are the stages of Colon Cancer ?

Staging evaluates how far cancer has spread through tissue, the lymph system, or the bloodstream using CT scans, MRI, PET scans, chest x-rays, surgery, lymph node biopsies, Complete Blood Counts (CBC), and CEA blood assays. The clinical stages include:

- **Stage 0 (Carcinoma in Situ):** Abnormal cells are found only in the innermost mucosal lining of the colon wall.
- **Stage I:** Cancer has formed in the mucosa and spread into the submucosa, and may have extended into the muscle layer of the colon wall. No lymph node involvement.
- **Stage II:** Subdivided into *Stage IIA* (spread through the muscle layer to the outer serosa), *Stage IIB* (spread through the serosa without invading nearby organs), and *Stage IIC* (spread through the serosa into nearby organs). No lymph node involvement.
- **Stage III:** Cancer involves regional lymph nodes:
 - *Stage IIIA:* Cancer involves the inner lining or muscle layer alongside 1 to 3 positive lymph nodes (or nearby tissue tumor cells), or involves the submucosa with 4 to 6 positive nodes.
 - *Stage IIIB:* Cancer involves outer layers without organ invasion alongside 1 to 3 positive nodes; or involves inner/muscle layers with 4 to 6 positive nodes; or involves mucosal/muscle layers with 7 or more positive nodes.
 - *Stage IIIC:* Cancer involves outer serosa with 4 to 6 positive nodes; or involves muscle/serosa layers with 7 or more positive nodes; or involves nearby organs with 1 or more positive nodes.
- **Stage IV:** Advanced metastatic disease:
 - *Stage IVA:* Spread to one distant organ not near the colon (e.g., liver, lung, ovary) or a distant lymph node.
 - *Stage IVB:* Spread to more than one distant organ or into the abdominal wall lining (peritoneum).

Question 7: What are the treatments for Colon Cancer ?

Management of colon cancer utilizes six standard modalities, often combined into multidisciplinary treatment plans based on stage and anatomical location:

- **Surgery:** The most common primary treatment for all stages. Options include *Local Excision / Polypectomy* for superficial early lesions; *Resection with Anastomosis* (partial colectomy removing the tumor, adjacent healthy tissue, and regional lymph nodes, followed by sewing healthy bowel ends together); and *Resection with Colostomy* (creating a temporary or permanent abdominal stoma when bowel ends cannot be rejoined). Post-operative chemotherapy or radiation given to eliminate residual cells and lower recurrence risk is called *adjuvant therapy*.
- **Radiofrequency Ablation (RFA):** Utilizes a special electrode probe inserted through the skin or via abdominal incision to kill localized cancer cells with high-energy radio waves.
- **Cryosurgery (Cryotherapy):** Deploys specialized instruments to freeze and destroy abnormal malignant tissue.
- **Chemotherapy:** Systemic or regional drugs designed to halt cancer cell division. For liver metastases, *chemoembolization of the hepatic artery* is often utilized to deliver concentrated chemotherapy directly to liver tumors while blocking their arterial blood supply.
- **Radiation Therapy:** Uses external radiation machines or internal radioactive seeds (brachytherapy) to destroy cancer cells. External radiation is frequently utilized as palliative therapy for advanced colon cancer to relieve symptoms and improve quality of life.
- **Targeted Therapy:** Drugs that attack specific cancer mechanisms without harming normal cells. Includes monoclonal antibodies (e.g., bevacizumab, ramucirumab, cetuximab, panitumumab) and angiogenesis inhibitors (e.g., ziv-aflibercept, regorafenib) that block tumor blood vessel growth or epidermal growth factor receptors.
- **Stage-Specific Management:**
 - *Stage 0 & I:* Treated primarily with surgical resection (local excision, polypectomy, or anastomosis).
 - *Stage II & III:* Managed via resection and anastomosis, frequently followed by adjuvant chemotherapy (standard for Stage III) or clinical trials.
 - *Stage IV & Recurrent:* Managed with local excision, wide resection of bowel and metastatic organs (e.g., liver, lungs, ovaries), pre/post-operative chemotherapy, RFA, cryosurgery, chemoembolization, palliative radiation/chemo, and targeted therapies.

Question 8: What research (or clinical trials) is being done for Colon Cancer ?

Clinical trials represent a vital avenue of cancer research, designed to scientifically evaluate whether new medications, drug combinations, or surgical procedures are safer and more effective than existing standard treatments.

Patients can enroll in clinical trials before starting initial therapy, during standard treatment protocols, or after standard regimens have proven ineffective. Participation helps advance clinical oncology and provides patients with access to emerging therapeutic modalities. Continued medical surveillance and follow-up testing remain mandatory for all patients to monitor treatment response and detect potential recurrence.

This document was generated for educational and clinical reference purposes based on official patient treatment guidelines (PDQ) from the National Cancer Institute.